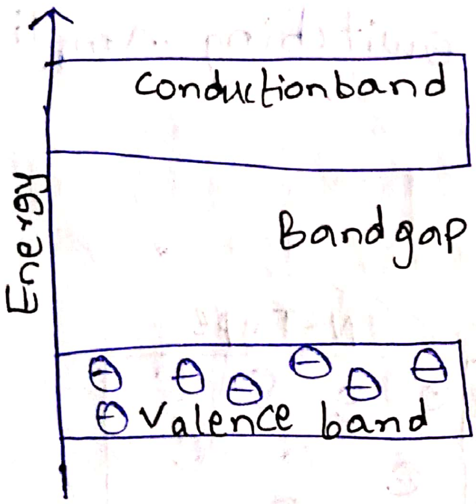
13/02/25

Semi Conductor

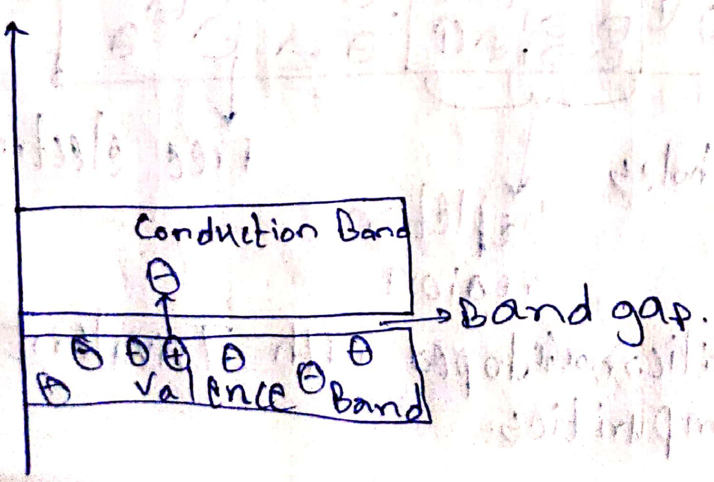
- Intrinsic
  - Pure form of semiconductor
  - Ex:- Pure form of Ge and Si.
- Extrensic
  - N-type
    - If we add penta-valent elements (or) impurities to intrinsic semiconductors
    - Ex:- Impurity P, As etc
  - P-type
    - If we add tri-valent impurities to intrinsic semiconductors.
    - Ex:- Ga, B, In

~~Inductor~~

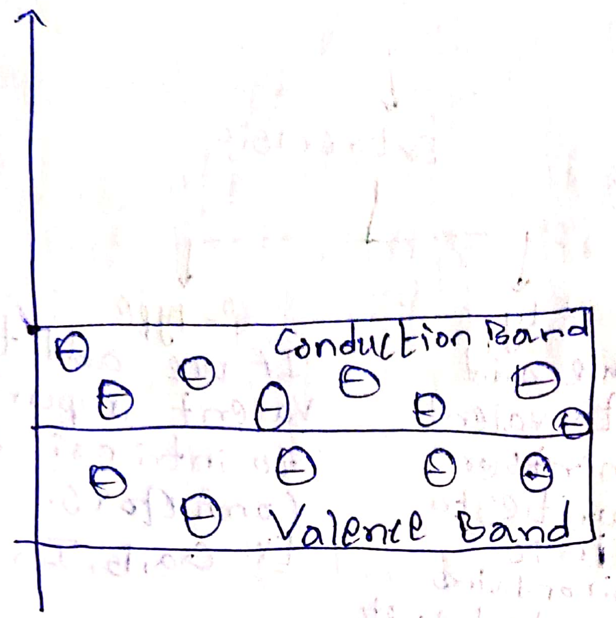
Insulator



Semi Conductor



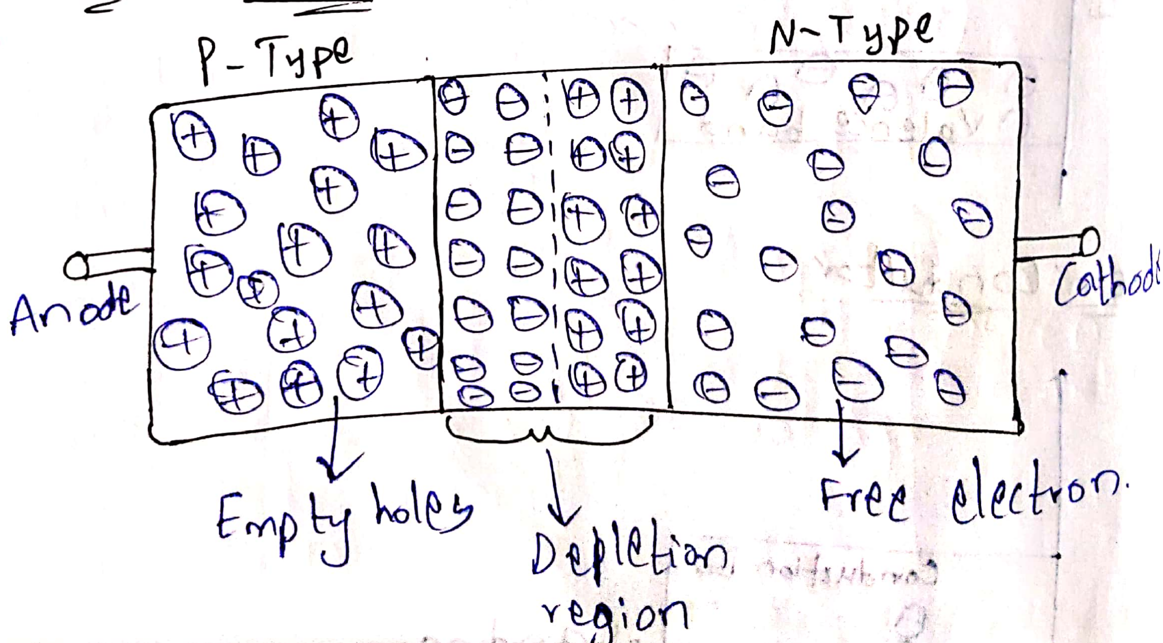
Conductor



* No band gap

* Semiconductors are used in the preparation of diodes, transistors & Integrated circuits (IC's).
* They are used for switching, amplification & signal processing.

PN Junction Diode

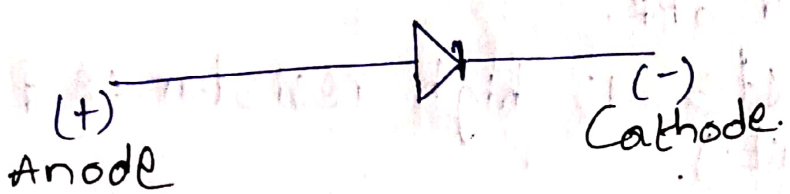


* P-Type : Silicon or the doped with Trivalent impurities.

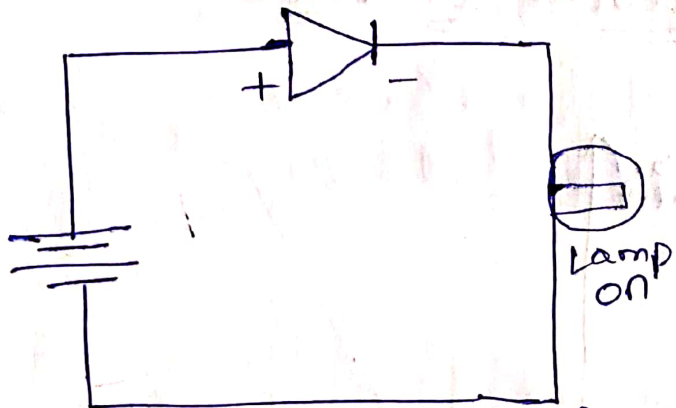
N-Type: Silicon (or) Ge doped with pentavalent impurities.

* A semiconductor diode that allows current flow in only one direction.

* Barrier potential for Silicon = 0.7V
  Germanium = 0.3V

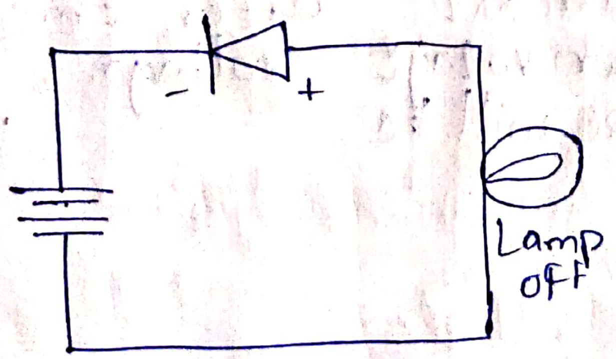


Forward Bias



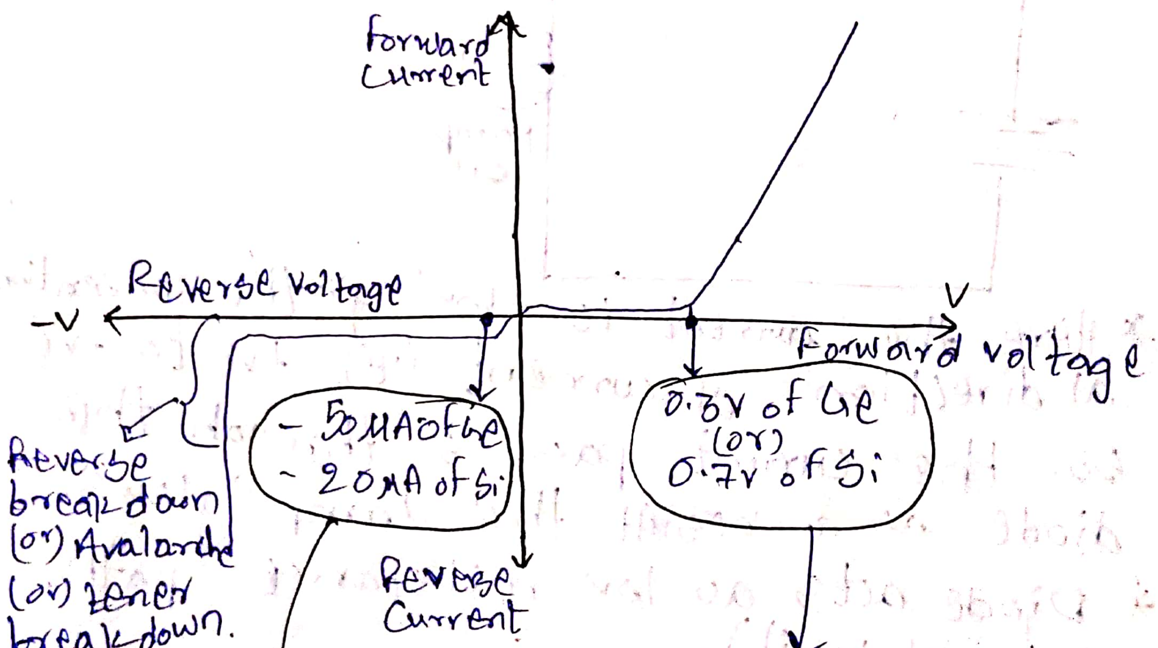
* Here the current is flowing in the conventional direction of current i.e; +ve to -ve so, the current passes through the diode as a result the lamp glows.
* Diode acts as low resistance path. (Closed circuit).

Reverse Bias



* Here the negative terminal of Diode & Positive terminal of Battery is connected & the positive side of diode & negative terminal of battery are connected, as a result there will be no passage of current through the diode so the lamp gets turned off.
* Diode acts as high resistance path (open circuit).

V-I characterstics of Diode



There will be minimal leakage of current until breakdown voltage after that there will be a drastic fall.

Rapid increase in the current after threshold voltage.

Diode Applications

* Diode acts as a switch.
* Rectification :- Converts AC to DC.
  Ex :- Full-wave Rectifiers & Half-wave Rectifiers.
* Clipping / Clamping :- Modifies waveform shapes.
* Voltage Regulation: zener diodes maintain constant voltage.
* Signal mixing / Demodulation : Used in communication circuits.

Diode current Equation

$$\boxed{I = I_0 \left(e^{\frac{qV}{\eta KT}} - 1 \right)}$$

I = current flowing through the diode.

I_0 = Saturation current.

q = Charge on the electron.

V = Voltage applied across the diode.

η = ideality factor.

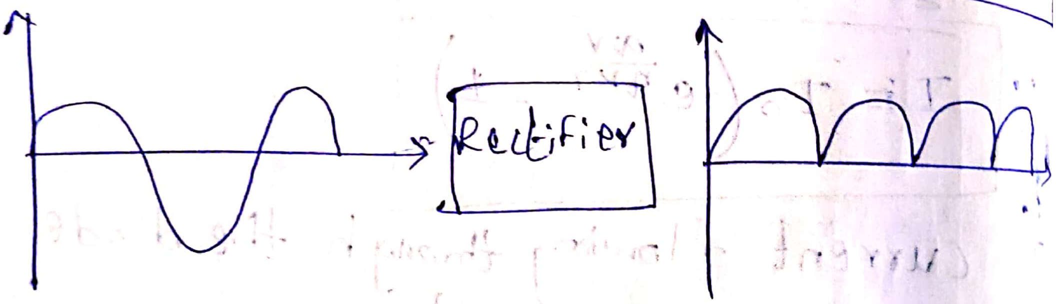
k = Boltzmann's constant $\boxed{k = 1.38 \times 10^{-23}\ JK^{-1}}$

T = Temperature in kelvin.
Absolute.

Rectifier

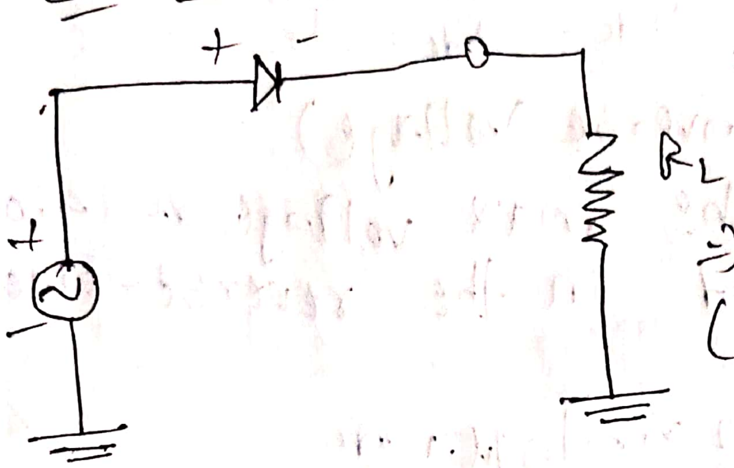
* It is an electronic device that convert AC into DC.
* It is used in electronic devices where DC power source is required.

Rectifier
- Half-wave
- Full-wave
  - Full-wave Centre tapped
  - Full-wave Bridge



→ I_{dc}, V_{dc}
→ I_{rms}, V_{rms}
→ γ = Ripple factor
→ P_{dc}, P_{ac} = Output & Input Power.
→ f = form factor
→ η = Efficiency (Ratio of Rectification).
→ Regulation.
→ Transformer Utilization Factor.
→ Peak Inverse Voltage.

Half-Wave Rectifier



⇒ Reverse Bias (Diode open circuit).
⇒ Forward Bias (Diode Closed circuit).

→ $I_{dc} = \frac{I_m}{\pi}$; $V_{dc} = \frac{V_m}{\pi}$

→ $I_{rms} = \frac{I_m}{2}$

→ $V_{rms} = \frac{V_m}{2}$

→ $\gamma = \sqrt{\left(\frac{I_{RMS}}{I_{dc}}\right)^2 - 1}$ ⇒ $\gamma = 1.211$.

* the ripple content in the output are 1.211 times the dc component i.e; 121.1 % of the dc component.

→ $\eta = \frac{\text{D.C output Power}}{\text{AC input Power}} = \frac{P_{dc}}{P_{ac}}$

$\eta = 40.6\%$ (H.W.R) $= \dfrac{\frac{I_m^2 R_L}{\pi^2}}{\frac{I_m^2}{4}(R_f + R_L + R_s)}$

→ Transformer Utilization Factor $= \frac{I_{DC}^2 R_L}{V_{rms} I_{rms}}$

* Ratio between the power delivered to the load and the ~~load~~ rated ac power from secondary.

$\boxed{TUF = 0.287}$

→ Form Factor = $\frac{I_{rms}}{I_{dc}} = \frac{V_{rms}}{V_{dc}} = 1.57$.

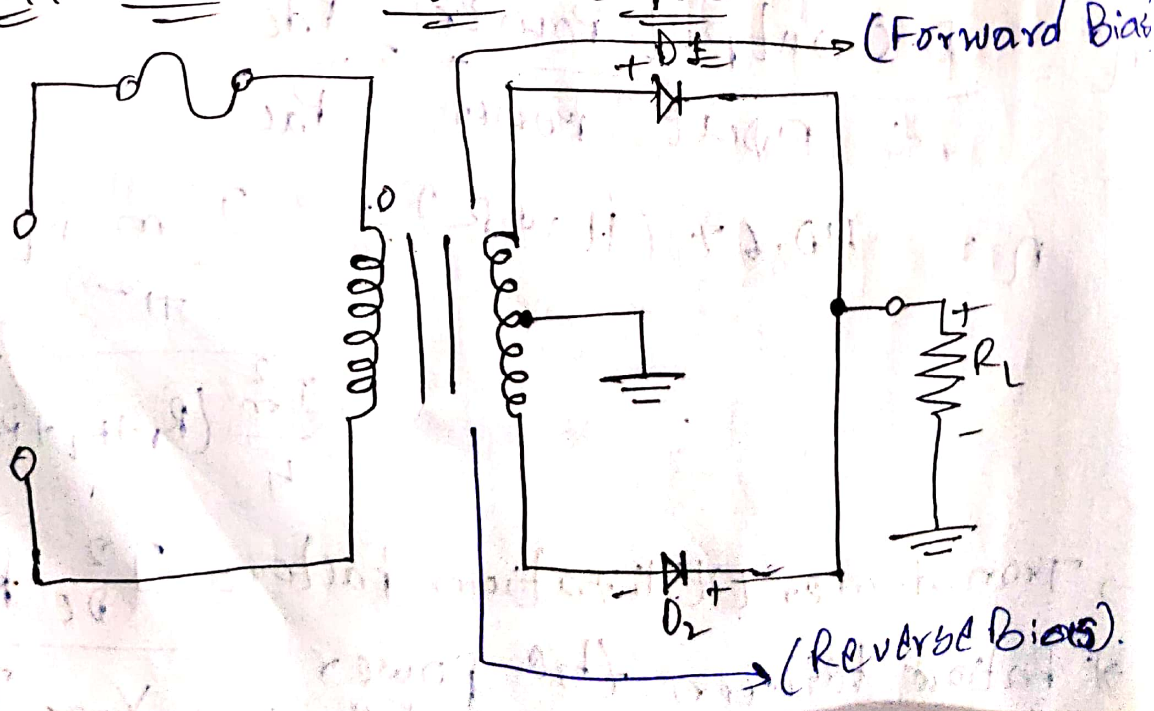
→ PIV (Peak Inverse Voltage)

* This is the max voltage a diode can withstand in the reverse-biased direction.

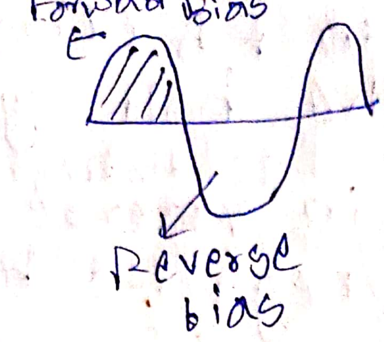
$\boxed{PIV = 2V_m}$ ⇒ H.W.R

* The only advantage of H.W.R is simple circuit & low cost.
* The disadvantages are:-
  → High ripple ~~per~~ content i.e; 121%
  → $\eta = 40.6\%$ which is inefficient.
  → TUF = 0.287 ; the transformer is not fully utilized.

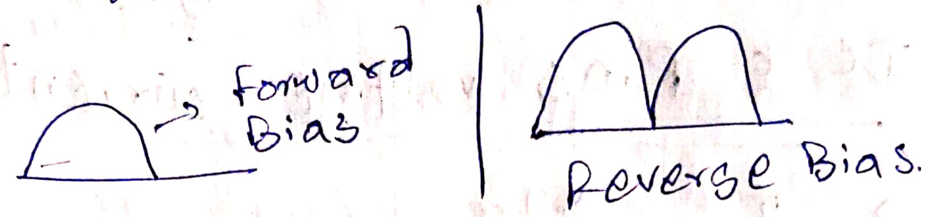
Full wave Center-tapped



Input:-



Output:-



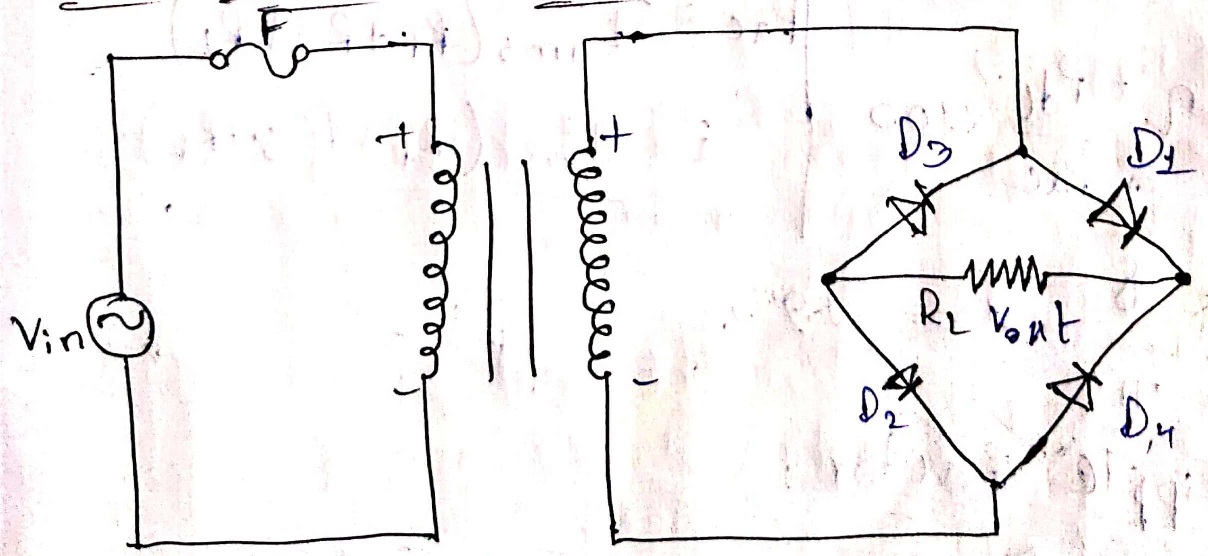
→ If D_1 is closed then it acts as forward Bias, else acts as reverse bias.

→ If D_2 is closed then it acts as forward

* If D1 is in F.B it is positive half cycle.

* Else D_2 is F.B it is negative half cycle.

Full wave Bridge Rectifier



* If D1 & D2 are in F.B (Closed circuit) then it is positive half cycle.

* If D3 & D4 are in ~~reverse~~ Forward bias (Closed ~~open~~ circuit) is negative half cycle.

DC & Rms voltage currents (F.W.R)

→ $I_{DC} = \frac{2I_m}{\pi}$

→ $I_{Rms} = \frac{I_m}{\sqrt{2}}$

→ $V_{DC} = \frac{2V_m}{\pi}$

→ $V_{Rms} = \frac{I_m}{\sqrt{2}} R_L$

* FWR efficiency

$\eta = \frac{P_{dc}}{P_{ac}}$

$P_{dc} = I^2_{dc} R_L = \frac{4}{\pi^2} I^2_m R_L$

$P_{ac} = I^2_{rms}(R_f + R_s + R_L)$

$= \frac{I^2_m (R_f + R_s + R_L)}{2}$

$\% \eta = \frac{P_{dc}}{P_{ac}} \times 100$

$= \frac{8}{\pi^2} \times 100 = 81.2\%$

Ripple Factor:

$\gamma = \sqrt{\left[\frac{I_{Rms}}{I_{DC}}\right]^2 - 1} = \sqrt{\frac{\pi^2}{8} - 1} = 0.48$

→ **TUF:-**

$$TUF = \frac{I_{DC}^2 R_L}{V_{rms} I_{rms}}$$

→ **PIV:-**

$$PIV = V_m$$

Parameter	H.W.R	F.W.R (C.T)	F.W.R (Bridge)
I_{dc}	I_m/π	$2I_m/\pi$	$2I_m/\pi$
V_{dc}	V_m/π	$2V_m/\pi$	$2V_m/\pi$
I_{rms}	$\frac{I_m}{2}$	$\frac{I_m}{\sqrt{2}}$	$\frac{I_m}{\sqrt{2}}$
V_{rms}	$\frac{V_m}{2}$	$\frac{V_m}{\sqrt{2}}$	$\frac{V_m}{\sqrt{2}}$
form factor	1.57	1.11	1.11
ripple factor	1.21	0.48	0.48
P_{dc}	$\left(\frac{I_m}{\pi}\right)^2 \times R_L$	$\left(\frac{2I_m}{\pi}\right)^2 \times R_L$	$\left(\frac{2I_m}{\pi}\right)^2 \times R_L$
P_{ac}	$I_m^2(R_s+R_f+R_L)/4$	$I_m^2(R_s+R_f+R_L)/2$	$I_m^2(R_s+R_f+R_L)/2$
η	40.6%	81.2%	81.2%
PIV	$2V_m$	V_m	V_m
TUF	0.287	$(0.574+0.812)/2$	0.812
%Reg	$(R_s+R_f)/R_L$	$(R_s/2+R_f)/R_L$	$(R_s+2R_f)/R_L$
No. of diodes	2	4	4
frequency	f	2f	2f

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Applications

→ Power supplies, Battery charging, Signal processing, Electrolysis, Renewable energy systems, Automotive applications.

Shape gets changed ← Wave Shaping Circuits

- Wave Shaping Circuits
  - Linear Wave Shaping
  - Non-Linear Wave Shaping
    - Clipper
    - Clamper

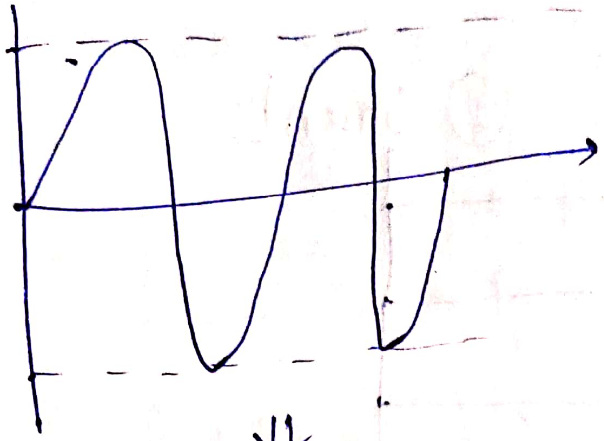
* Non-linear wave shaping circuits utili diodes alongside resistors to modify waveforms.

Clippers

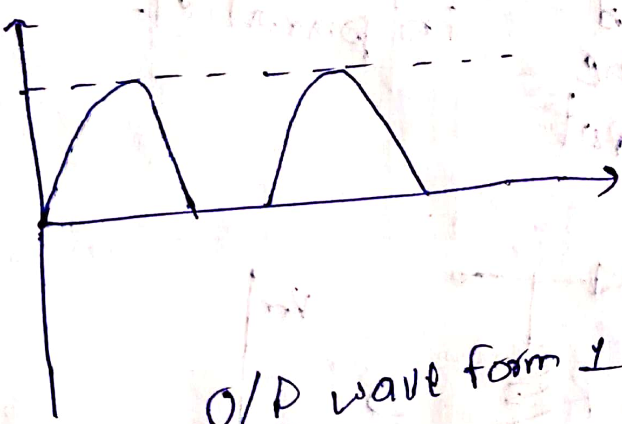
* A non-linear wave shaping circuit which can cut or trim the unwanted portio of the wave form.

* It prevents the out put waveform from exceeding a certain level, at the same time it does not distort the remaining part of the waveform.

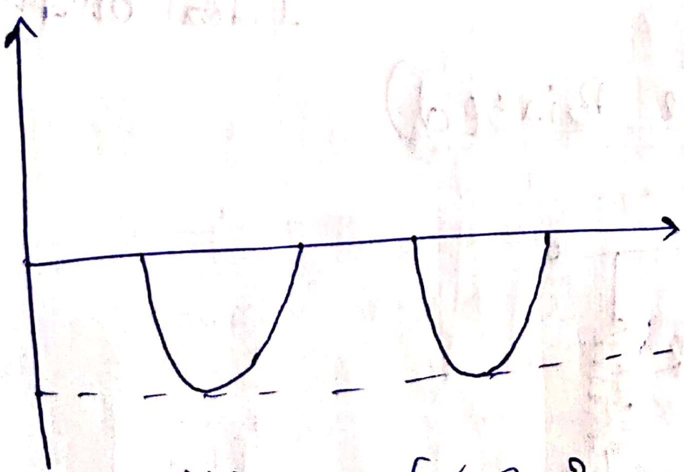
Ex:-



CLIPPER CIRCUIT



O/P wave form 1



O/P wave form 2

Types

(1) Series

- Positive
- Negative
- Positive Biased
- Negative Biased

⇓

Diode is connected in series with the load resistor & input signal.

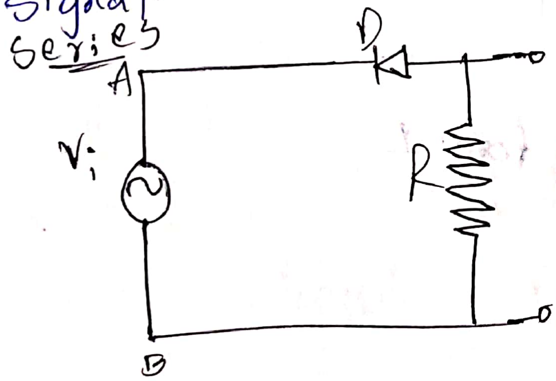
(2) Shunt

- Positive
- Negative
- Positive Biased
- Negative Biased

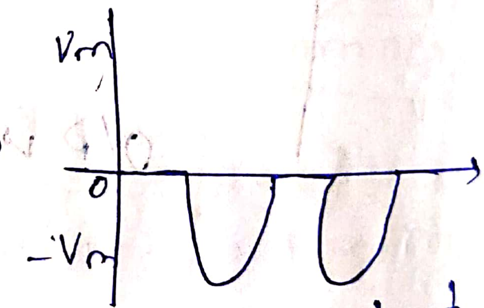
⇓

Diode is connected in parallel.

Series

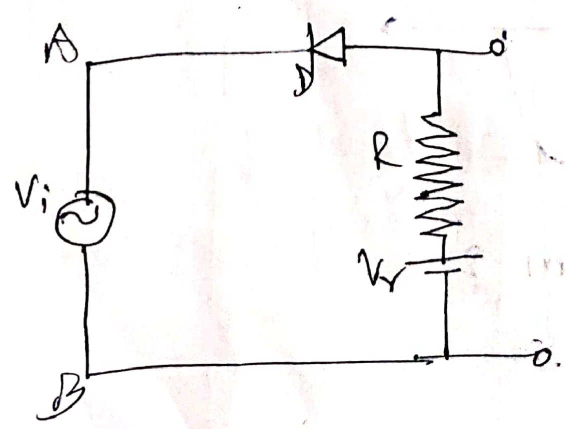


⇒

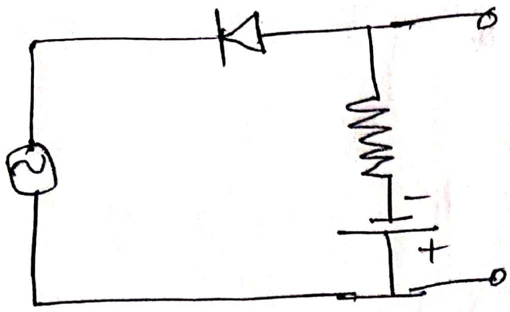


Ideal output.

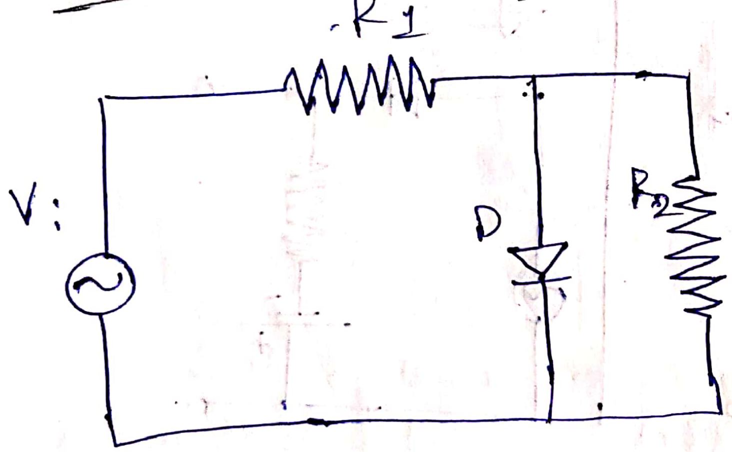
Series (Positive Biased)



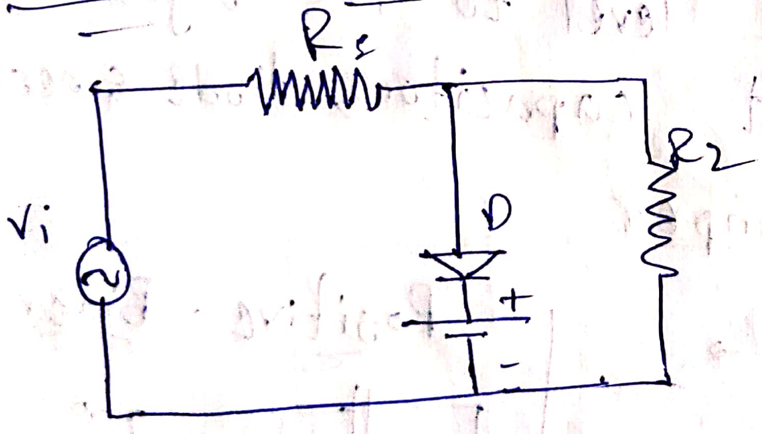
Series (Negative Biased)



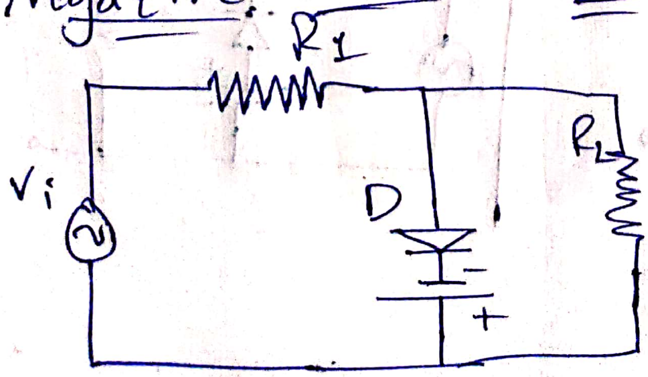
Positive Shunt Clipper



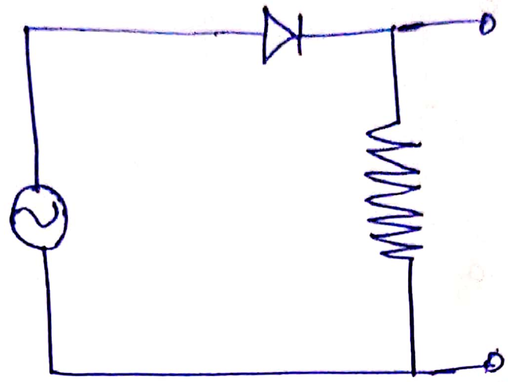
⇒ Positive Biased Shunt Clipper



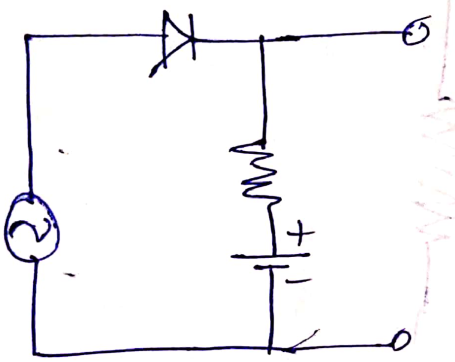
⇒ Negative Biased Shunt Clipper

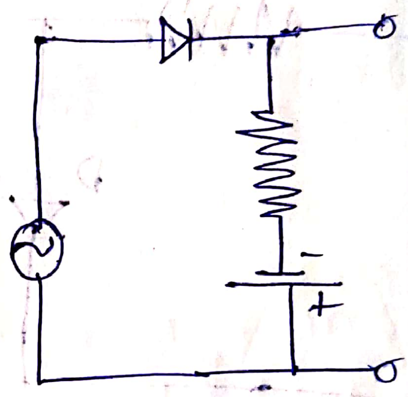


Negative Clipper



Positive Biased | Negative Biased



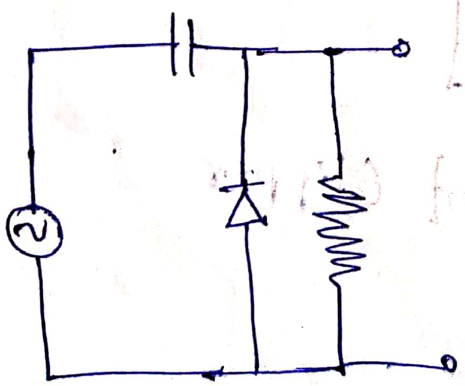


Clampers

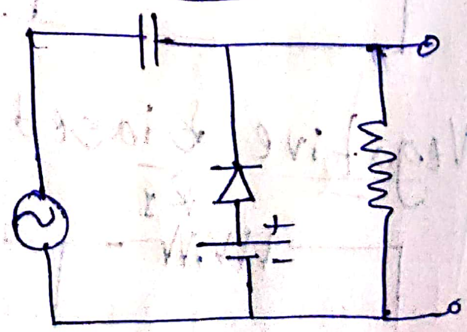
→ Adds a DC level to AC signal.

→ They consist capacitors, diode & resistor

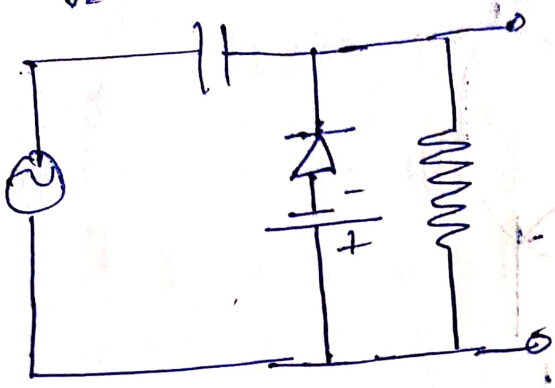
Positive clamper



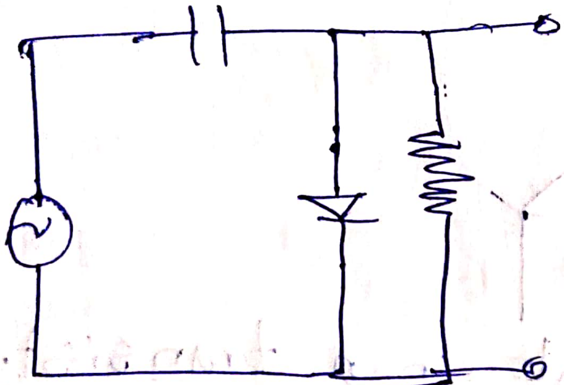
Positive -Biased



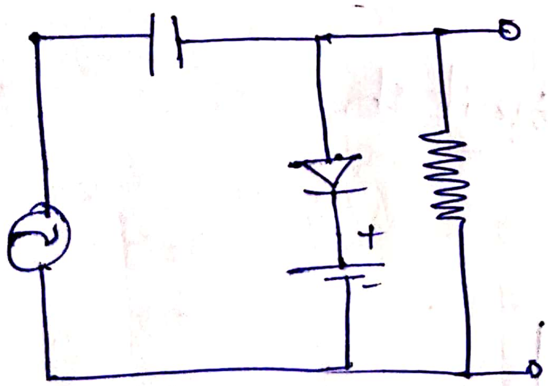
Negative - Biased



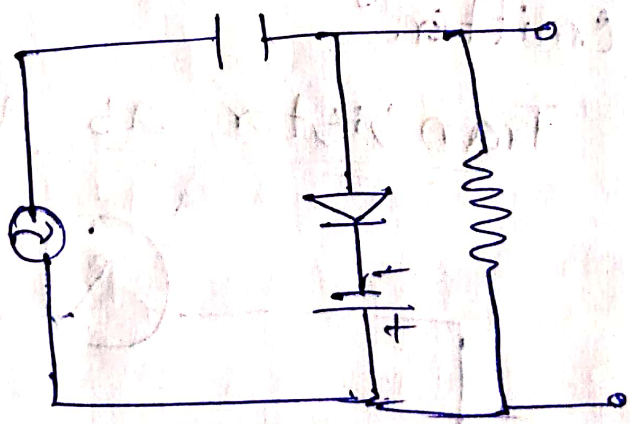
Negative Clamper



Positive - Biased)

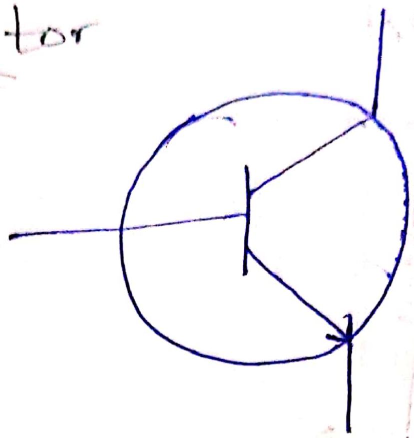


(Negative - Biased)

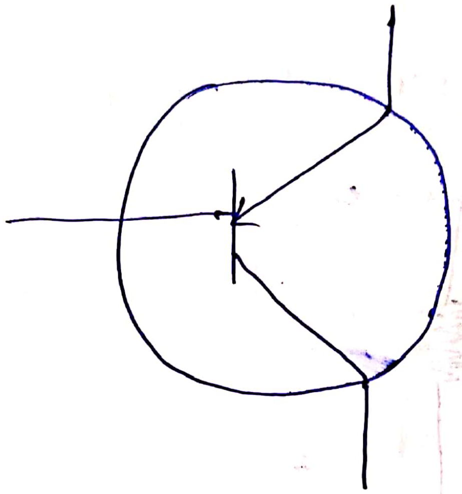


Transistor

NPN →

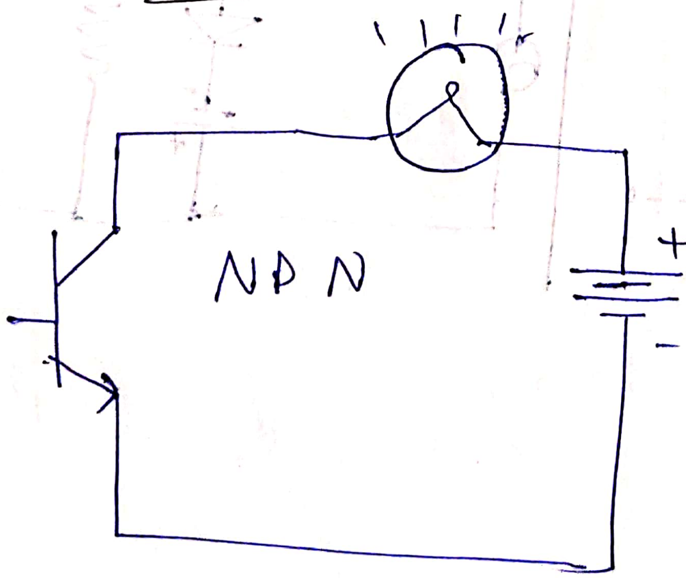


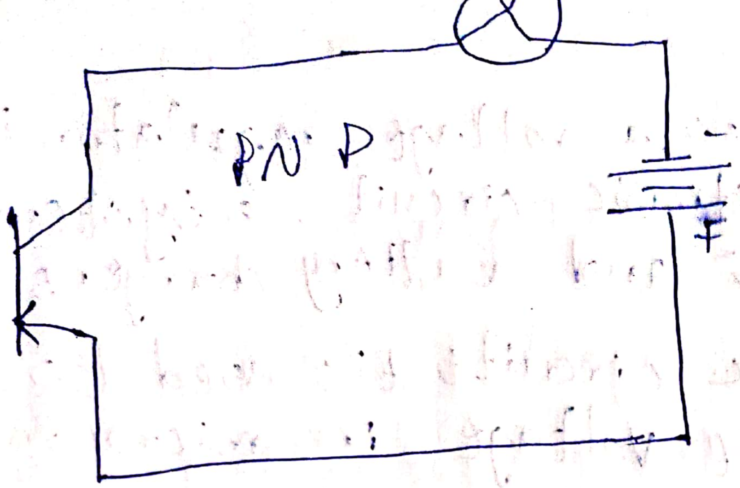
PNP →



* Controlled current in a transistor should go between the collector & the emitter.

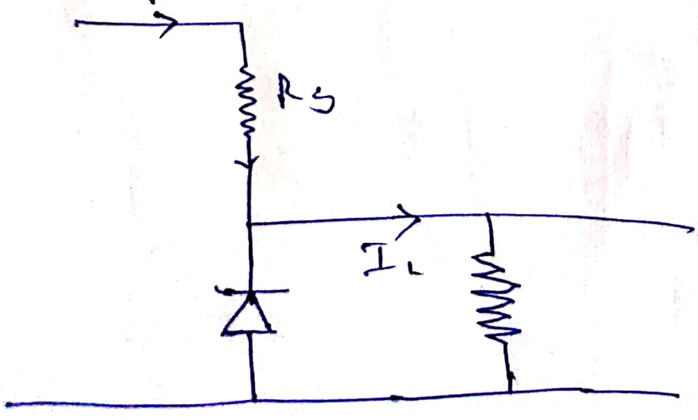
Transistor as a switch





* When a transistor has maximum current through it, it is said to be in a state of saturation.(Fully conducting).
* When a transistor has zero current through it, it is in a state of cutoff (non - conducting).

Zener Diode
→ It is a semiconductor device that allows current to flow in the reverse direction when a certain voltage is applied, efficitively acting as voltage regulator.
→ When voltage exceeds breakdown voltage diode begins to conduct & regulate the voltage.



⇒ zener diode acts as voltage regulator.

Applications

* zener diode as a voltage regulator is used in electronic circuit & system, power supplies and Battery chargers.

* In electronic circuits it is used to regulate the voltage in microcont sensors, amplifiers.

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CO-4

INTEGRATED CIRCUIT

→ In an IC millions of resistors, transistors & capacitors are fabricated.

→ multiple electronic components are fabricated on a single chip of semiconductor material then it is an IC.

Advantages

→ Compact size
→ Lesser weight
→ Low power consumption
→ Reduced cost
→ Increased reliability.
→ Better operating speeds.

Types

① Analog Integrated Circuits

→ IC's that operate over an entire range of continuous values of signal amplitude are called as Analog IC's.

Ex:- Op-amps, voltage regulators & audio amplifiers.

② Digital IC's

→ IC's that process discrete signals or information represented as a sequence of numbers.

Ex:- microprocessors, memory chips & logic gates.

Characterstics of Analog IC's

1. Continuous signal processing.
2. Signal Accuracy & Precision.
3. Variable Voltage levels
4. Complexity & precision in Design.
5. Low Power consumption.

Characterstics of Digital IC's

1. Binary signal Processing
2. Boolean logic operations.
3. Robustness to Noise
4. Design Scalability.
5. Power Consumption.
6. Data storage & Processing.

* Analog IC's consist less number of transistors compared to digital IC's

Voltage Regulators

* Voltage Regulator is an electronic device that maintains a constant output voltage level.

* It ensures a stable & reliable power supply for electronic devices.

*As the variation in voltage in electronic devices leads to malfunctions. So, we use voltage Regulators.

Types (According to working Principle).

1. Linear Voltage Regulators

→ Uses a series pass transistor to adjust the output voltage.

→ Limited efficiency.

2. Switching Voltage Regulators

→ These regulators switch the input voltage ON & OFF rapidly & use inductors & Capacitors to filter the output.

→ Provide higher efficiency.

→ According to voltage generated.

1. Positive Voltage Regulators

* Positive voltage.

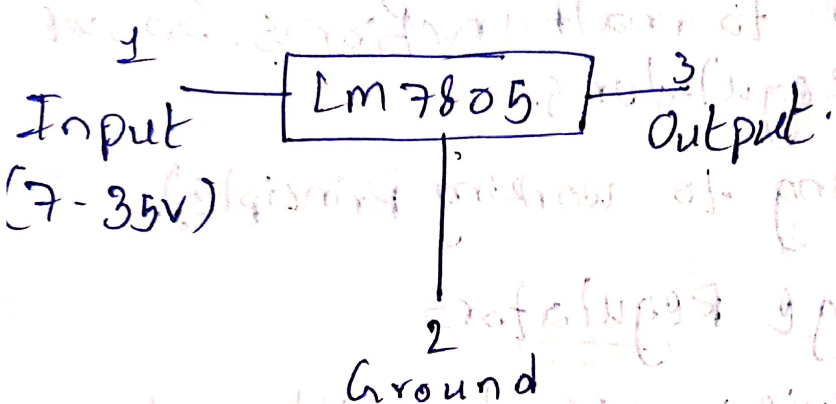
Ex:- LM 78XX series.

② Negative V.R

* Negative voltage

Ex:- LM 79XX.

LM 7805

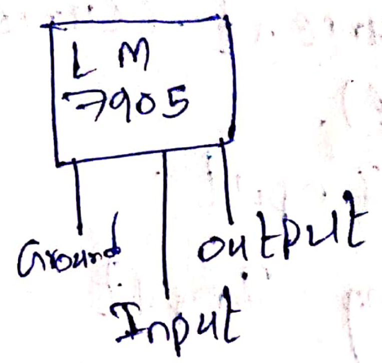


* Max efficiency for an input is 7.2V
* Fixed positive output voltage is 5V.
* Other variants include LM7805, LM7805C, LM7805CT which can handle upto 1A.

Applications

→ Current regulator.
→ Regulated dual supply.
→ Building circuits for Phone Charger, UPS power supply circuits, portable CD player.
→ Adjustable output regulator.
→ Fixed output regulator.

LM 7905



* Receives the regulated output of -5V.

* Input voltage range is -7V to -25V.

* Other variants include LM7905, LM7905 L

- LM7905 CT upto 1A.

Applications

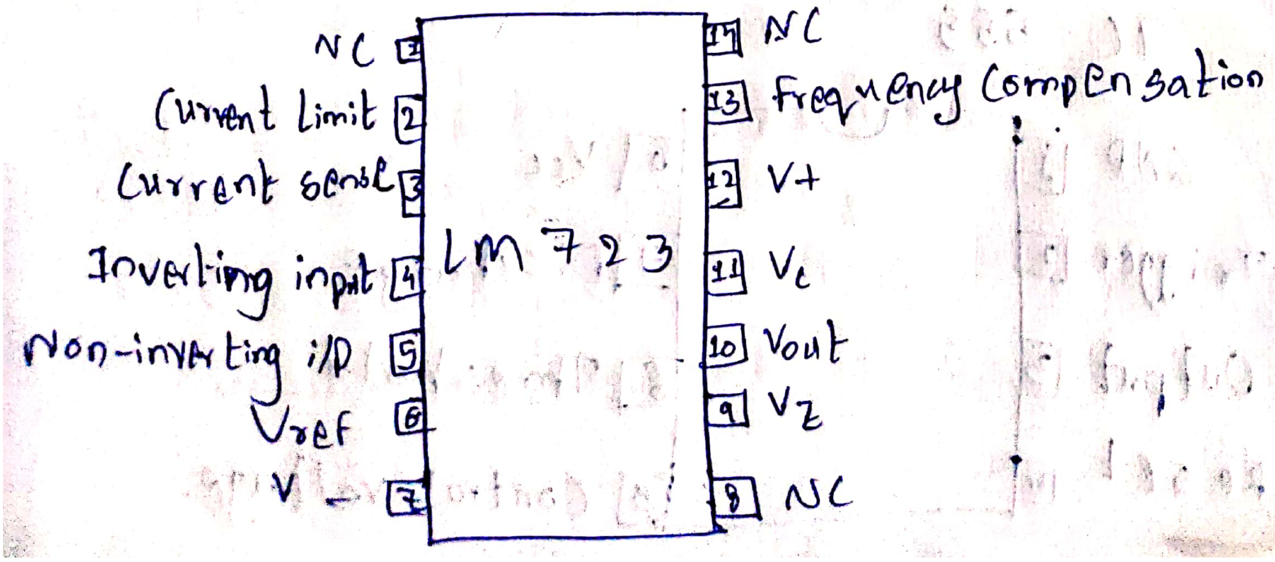
→ Used for op-amp signals.

→ Regulated Dual supply.

→ Limits current in certain applications.

LM 723

→ It is a changeable voltage regulator used in the series regulator application with 150mA current output without exterior pass transistor.



Pin 1 :- Not Connected.

Pin 2 :- Limits the current. (Current li

Pin 3 :- (Current sense).

Pin 4 :- Inverting Input.

Pin 5 :- Non-Inverting Input

Pin 6 :- provides 7V reference voltag

Pin 7 :- GND

Pin 8 :- NC

Pin 9 :- This pin used to make negative regulators.

Pin 10 :- O/P pin.

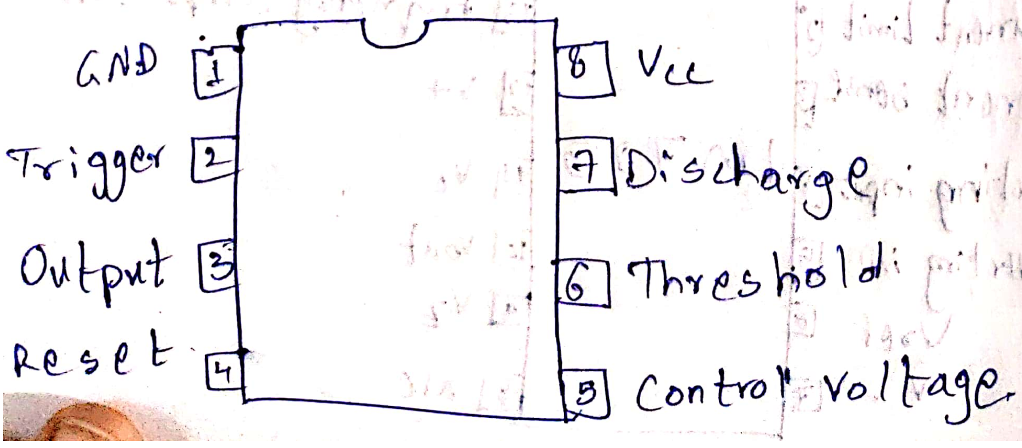
Pin 11 :- This is series pass transist collector input.

Pin 12 :- Positive Supply.

Pin 13 :- Decreases noise with a 100 pF capacitor.

Pin 14 :- NC

IC 555



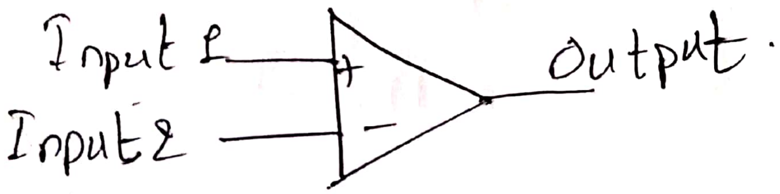
Applications

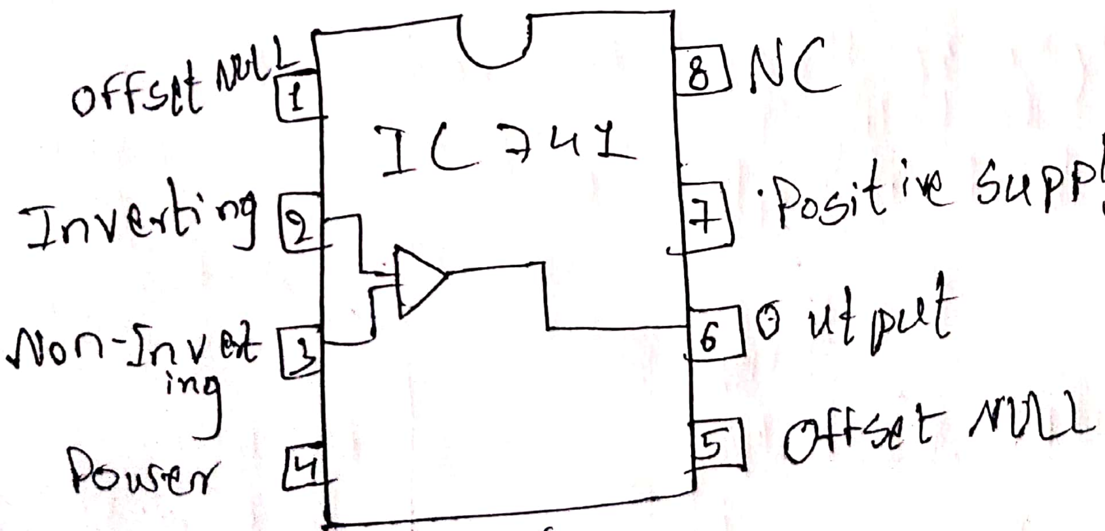
→ Timers
→ Dimmer circuits
→ PULSE Generator circuits.
→ Square wave generators.
→ Duty cycle adjuster.

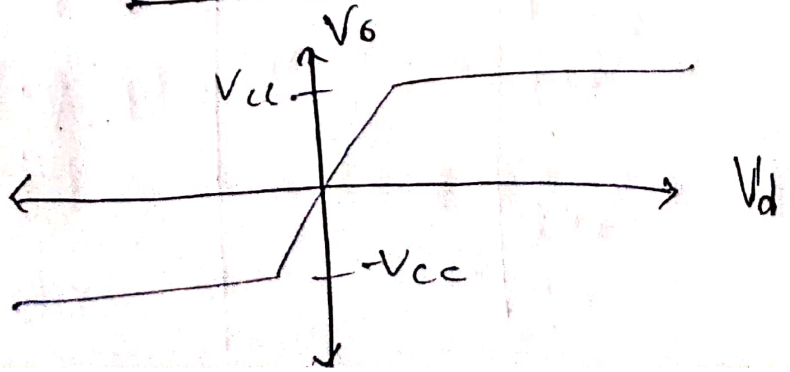
Operational Amplifier

→ Op amp performs mathematical operations by using analog signals.

→ Used to build many analog circuits, filters, oscillators, amplifiers.







Applications

→ Voltage follower
→ Inverting amplifier.
→ Non-inverting amplifier.
→ Summing amplifier.
→ Differential amplifier
→ Integrator